

An Analysis of Question answering

Question answering

Question answering -- Part 1

In information retrieval, an open-domain question answering system tries to return an answer in response to the user's question. The returned answer is in the form of short texts rather than a list of relevant documents. The system finds answers by using a combination of techniques from computational linguistics, information retrieval, and knowledge representation. The system takes a natural language question as an input rather than a set of keywords, for example: "When is the national day of China?" It then transforms this input sentence into a query in its logical form. Accepting natural language questions makes the system more user-friendly, but harder to implement, as there are a variety of question types and the system will have to identify the correct one in order to give a sensible answer. Assigning a question type to the question is a crucial task; the entire answer extraction process relies on finding the correct question type and hence the correct answer type. Keyword extraction is the first step in identifying the input question type. In some cases, words clearly indicate the question type, e.g., "Who", "Where", "When", or "How many"—these words might suggest to the system that the answers should be of type "Person", "Location", "Date", or "Number", respectively. POS (part-of-speech) tagging and syntactic parsing techniques can also determine the answer type. In the example above, the subject is "Chinese National Day", the predicate is "is" and the adverbial modifier is "when", therefore the answer type is "Date". Unfortunately, some interrogative words like "Which", "What", or "How" do not correspond to unambiguous answer types: Each can represent more than one type. In situations like this, other words in the question need to be considered. A lexical dictionary such as WordNet can be used for understanding the context. Once the system identifies the question type, it uses an information retrieval system to find a set of documents that contain the correct keywords. A tagger and NP/Verb Group chunker can verify whether the correct entities and relations are mentioned in the found documents. For questions such as "Who" or "Where", a named-entity recogniser finds relevant "Person" and "Location" names from the retrieved documents. Only the relevant paragraphs are selected for ranking. A vector space model can classify the candidate answers. Check if the answer is of the correct type as determined in the question type analysis stage. An inference technique can validate the candidate answers. A score is then given to each of these candidates according to the number of question words it contains and how close these words are to the candidate—the more and the closer the better. The answer is then translated by parsing into a compact and meaningful representation. In the previous example, the expected output answer is "1st Oct."

Question answering -- Part 2

Methods Question answering is dependent on a good search corpus; without documents containing the answer, there is little any question answering system can do. Larger collections generally mean better question answering performance, unless the question domain is orthogonal to the collection. Data redundancy in massive collections, such as the web, means that nuggets of information are likely to be phrased in many different ways in differing contexts and documents, leading to two benefits:

Question answering -- Part 3

Further reading Dragomir R. Radev, John Prager, and Valerie Samn. Ranking suspected answers to natural language questions using predictive annotation Archived 2011-08-26 at the Wayback Machine. In Proceedings of the 6th Conference on Applied Natural Language Processing, Seattle, WA, May 2000. John Prager, Eric Brown, Anni Coden, and Dragomir Radev. Question-answering by predictive annotation Archived 2011-08-23 at the Wayback Machine. In Proceedings, 23rd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Athens, Greece, July 2000. Hutchins, W. John; Harold L. Somers (1992). An Introduction to Machine Translation. London: Academic Press. ISBN 978-0-12-362830-5. L. Fortnow, Steve Homer (2002/2003). A Short History of Computational Complexity. In D. van Dalen, J. Dawson, and A. Kanamori, editors, The History of Mathematical Logic. North-Holland, Amsterdam. Tunstall, Lewis (5 July 2022). Natural Language Processing with Transformers: Building Language Applications with Hugging Face (2nd ed.). O'Reilly UK Ltd. p. Chapter 7. ISBN 978-1098136796.

Question answering -- Part 4

If the right information appears in many forms, the question answering system needs to perform fewer complex NLP techniques to understand the text. Correct answers can be filtered from false positives because the system can rely on versions of the correct answer appearing more times in the corpus than incorrect ones. Some question answering systems rely heavily on automated reasoning.

Question answering -- Part 5

Answering questions related to an article in order to evaluate reading comprehension is one of the simpler form of question answering, since a given article is relatively short compared to the domains of other types of question-answering problems. An

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example of such a question is "What did Albert Einstein win the Nobel Prize for?" after an article about this subject is given to the system. Closed-book question answering is when a system has memorized some facts during training and can answer questions without explicitly being given a context. This is similar to humans taking closed-book exams. Closed-domain question answering deals with questions under a specific domain (for example, medicine or automotive maintenance) and can exploit domain-specific knowledge frequently formalized in ontologies. Alternatively, "closed-domain" might refer to a situation where only a limited type of questions are accepted, such as questions asking for descriptive rather than procedural information. Question answering systems in the context of machine reading applications have also been constructed in the medical domain, for instance related to Alzheimer's disease. Open-domain question answering deals with questions about nearly anything and can only rely on general ontologies and world knowledge. Systems designed for open-domain question answering usually have much more data available from which to extract the answer. An example of an open-domain question is "What did Albert Einstein win the Nobel Prize for?" while no article about this subject is given to the system. Another way to categorize question-answering systems is by the technical approach used. There are a number of different types of QA systems, including:

Question answering -- Part 6

Question answering (QA) is a computer science discipline within the fields of information retrieval and natural language processing (NLP) that is concerned with building systems that automatically answer questions that are posed by humans in a natural language. A question-answering implementation, usually a computer program, may construct its answers by querying a structured database of knowledge or information, usually a knowledge base. More commonly, question-answering systems can pull answers from an unstructured collection of natural language documents. Some examples of natural language document collections used for question answering systems include reference texts, compiled newswire reports, Wikipedia pages and other World Wide Web pages.